

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

May 2014

CL316 Highway Engineering

Date: 06.05.2014, Tuesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Use of IRC charts is allowed.

SECTION - I

- Q - 1 (a) Draw a typical cross section of MDR in cutting in Rural area. [02]
- (b) A vertical summit curve is formed when ascending gradient of 1 in 25 m another descending gradient of 1 in 100. Find the length of summit curve to provide the required stopping distance for a design speed of 80 kmph. [05]
- Q - 2 (a) Write a note on highway financing. [05]
- (b) A national highway passing through a flat terrain has a horizontal curve of radius equal to ruling minimum radius. If the design speed is 100 kmph, calculate absolute minimum sight distance, superelevation, extra widening and length of transition curve. Assume necessary data suitably. [05]
- (c) What are the advantages and drawbacks of cement concrete roads? Explain cement grouted and roller concrete layers and their uses. [04]

OR

- Q - 2 (a) Explain briefly various factors affecting the vehicle operation cost. [05]
- (b) The speeds of overtaking and overtaken vehicles are 70 and 40 kmph respectively on a two way traffic road. If the acceleration of overtaking vehicle is 0.99 m/sec^2 . (i) Calculate safe overtaking sight distance, (ii) mention the minimum length of overtaking zone and (iii) draw a neat sketch of overtaking zone and show the position of the sign posts. [05]
- (c) Discuss the objects of following types of joints with neat sketches: [04]
- (i) Expansion joints
 - (ii) Longitudinal joints
- Q - 3 (a) Write construction steps for WBM road. [02]
- (b) A cc pavement slab of thickness 20 cm is constructed over a sub base having modulus of reaction 15 kg/cm^3 . The temperature difference is found to be 18°C . The spacing [06]

between transverse joint is 4.5 m and longitudinal joint is 3.5 m. the design wheel load is 5100 kg, radius of contact area is 15 cm. find the worst combination of stresses at the edge. Take $E = 3 \times 10^5 \text{ kg/cm}^2$, poisson's ratio 0.15, co-efficient of thermal expansion 10×10^{-6} per $^{\circ}\text{C}$ and friction coefficient is 1.5.

- (c) Discuss relationship between tyre pressure and contact pressure. [06]

OR

- Q - 3 (a) Explain seal coat and penetration macadam. [02]

- (b) Design the size and spacing of dowel bars at the expansion joints of a cement concrete pavement thickness 25 cm with radius of relative stiffness 80 cm. for a design wheel load of 5000 kg. Assume load capacity of dowel system as 40% of the design wheel load. Joint width is 2 cm, permissible shear and flexural stresses in dowel bar are 1000 and 1400 kg/cm^2 and permissible bearing stress in CC is 100 kg/cm^2 . [06]

- (c) A sub-grade soil sample has the following properties: [06]

Soil passing 75 micron sieve = 60%

Liquid limit = 65 %

Plastic limit = 45%

Design the pavement section by group index method for heavy traffic with over 400 commercial vehicles per day. What are the limitations of this method?

SECTION - II

- Q - 4 (a) Define: Traffic volume, Traffic density [02]

- (b) Write procedure to find spot speed study. [05]

- Q - 5 (a) Draw a neat sketch of following traffic signs: [05]

(i) Stop, (ii) restriction end, (iii) give way, (iv) load limit, (v) Axle load limit

- (b) Define subsurface drainage and explain procedure for lowering the water table. [05]

- (c) Enlist different failures in flexible pavement. [04]

OR

- Q - 5 (a) Draw a neat sketch of following traffic signs: [05]

(i) Narrow bridge (ii) T-intersection (iii) staggered intersection, (iv) loose gravel, (v) side road right

- (b) Enlist importance of highway drainage and requirements of it. [05]

- (c) Explain consolidation of flexible pavement layer. [04]

- Q - 6 (a) Give the names of drawings to be prepared for finalizing the alignment. [04]

- (b) How macadam's construction is better than the other types of construction. [05]

- (c) What are the steps to follow to find the toughness of the aggregates? [05]

OR

Q - 6 (a) How obligatory points through centre line should not pass, affect the alignment? [04]

(b) Calculate the road length of a district as per Nagpur road plan. [05]

(i) Total area = 6300 km²

(ii) length of railway track = 75 km

(iii) Agricultural area = 2800 km²

No. of villages with population ranges <500, 501-100, 1001-2000, 2001-5000 and above 5001 are 450, 320, 110, 50 and 10 respectively.

Population range	10001 – 20000	20001 – 50000	50001 – 100000	>100000
No. of towns	20	10	6	2

(c) Explain float test. [05]

$$NH + SH + MDR \text{ (km)} = [A/8 + B/32 + 1.6N + 8T] + D - R$$

$$ODR + VR \text{ (km)} = [0.32V + 0.8Q + 1.6P + 3.2S] + D$$

$$l = \left[\frac{Eh^3}{12K(1-\mu^2)} \right]^{\frac{1}{4}} \quad S_{ts} = \frac{EetC_s}{2} \quad S_f = \frac{WL_x f}{2 \times 10^4} \quad S_{ts} = \frac{Eet}{3(1-\mu)} \quad , Gl = 0.2a + 0.005 ac + 0.01 bd$$

$$P' = 0.785 d^2 F_z \quad P' = \frac{2d^2 F_f}{L_d + 8.8 \delta} \quad P' = \frac{F_b L_d^2 d}{12.5 (L_d + 1.5 \delta)} \quad L_d = 5d \left[\frac{F_f}{F_b} \times \frac{L_d + 1.5 \delta}{L_d + 8.8 \delta} \right]^{\frac{1}{2}}$$